

MAPPING OYSTER REEF HABITAT IN SOUTH CAROLINA

A newly developed intertidal oyster reef and elevational suitability map for Folly Island, to help support resilience planning and wild oyster harvest in South Carolina.

CORE SC 2024-25 RESEARCH PROJECT

PI: Susan Wang, Professor of Geography, University of South Carolina (USC).

Co-PI: Gary Sundin, S.C. Department of Natural Resources (SCDNR).

Collaborators:

- Caitlyn Bierce, College of Charleston.
- Sarah Pedigo, S.C. Sea Grant Consortium.
- Matt Gorstein, S.C. Sea Grant Consortium.

METHODS

- 2020 multispectral imagery provided through the South Carolina Aerial Imagery Program (Geographic Information Council).
- 2024-25 Ground truthing drone flights provided by SCDNR.

A Deep Learning approach was used to map oyster reefs and marsh (both submerged and not submerged) at a precision of 83.63%

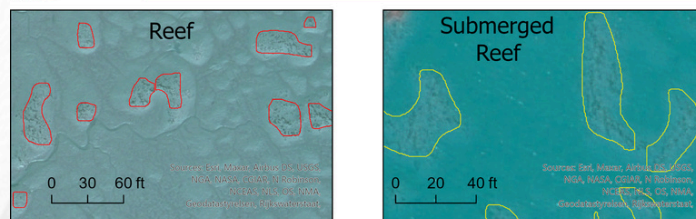
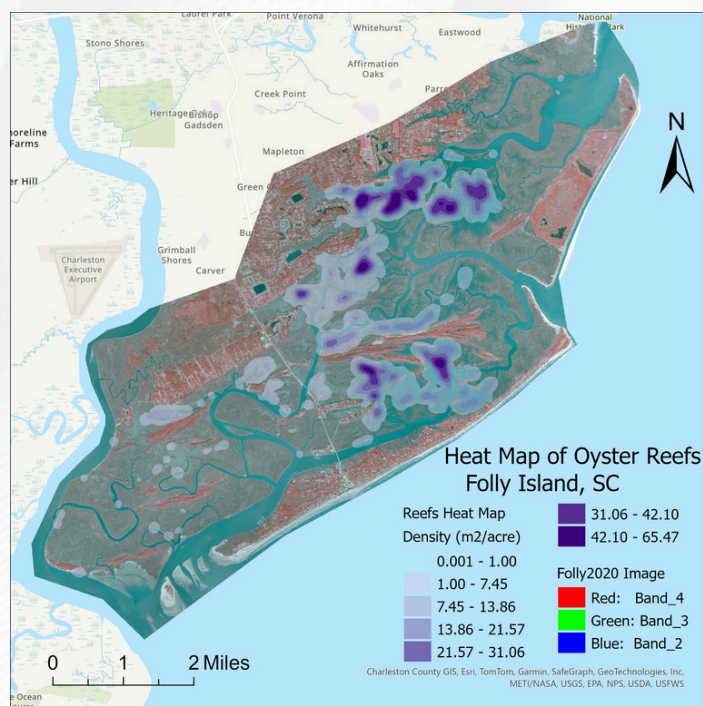
The current elevation of reefs was used to create an elevation suitability index.

The 2022 NOAA Sea Level Rise Technical Report projects a 10-25" rise in mean sea level along the U.S. coastline by 2050.

PROJECT OBJECTIVES

Eastern oysters (*Crassostrea virginica*) are the 4th largest commercial fishery product in S.C. with wild oyster reefs growing throughout the intertidal zone providing valuable ecosystem services.

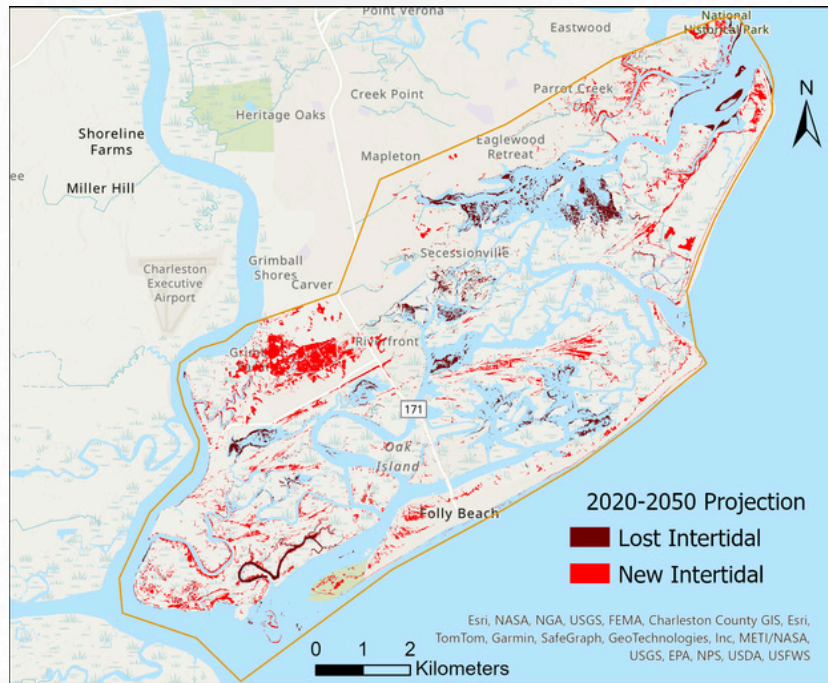
- Map the location of current oyster reefs in 2020 imagery.
- Develop an Elevational Suitability Index to predict future reef suitability.



A heat map showing oyster reef clustering at lower elevation in 2020.

OYSTER SUITABILITY MODELING

An Elevational Habitat Suitability Index is established to assist oyster managers and industry practitioners, helping direct their harvest, husbandry, and restoration efforts.
(Dr. Susan Wang, USC Geography)



2020-2050 Projected Tidal Land Change on Folly Beach.

SEA LEVEL RISE SCENARIO FOLLY BEACH, S.C.

According to their latest Adaptation Plan, the City of Folly Beach is planning for 1 foot of sea level rise by 2050.

Projected Intertidal Land Change:

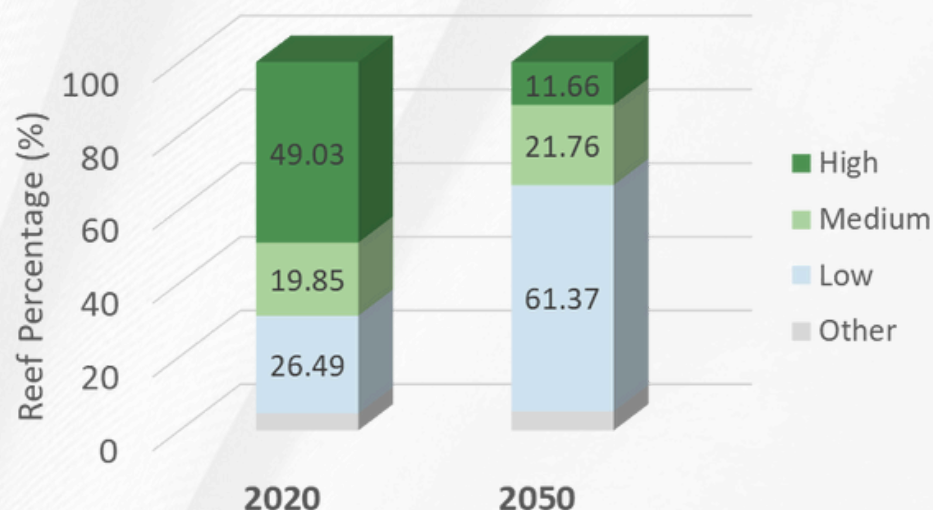
- New Subtidal (Former Inter): 713.89 acres
- New Intertidal 1,305.13 acres.

Current oyster reefs around Folly Island are clustered in lower elevations making them especially vulnerable to sea level rise.

In a 1-ft sea level rise scenario, by 2050 current reefs at Folly Island will remain, but habitat suitability will be heavily affected:

- 97% high ranking reefs downgraded
- 74% medium ranking reefs downgraded
- 61% of current reefs will be in LOW suitability by 2050.

OYSTER HABITAT SUITABILITY CHANGE MATRIX



2020-2050 change in suitability ranks of current oyster reefs

MAPPING FOLLY ISLAND'S OYSTER REEF HABITAT

April 9, 2025

SCDRN MRRI Auditorium
217 Fort Johnson Road
Charleston, SC 29412

Agenda

10:00-10:30am | Registration

10:30-10:35am | Opening remarks with *Gary Sundin* (SCDNR MRRI)

**10:35-10:45am | *Susan Anderson* CORE SC
(Assistant Vice President for Research, CofC)**

**10:45-10:55am | *Matthew Gorstein* SC Sea Grant Consortium
(Director for Development and Extension)**

**10:55-11:05am | *Matthew Wellslager* SC Aerial Program
Initiative, SC Revenue and Fiscal Affairs Office (Program
Manager)**

11:05-11:50am | Project Report with *Susan Wang* (USC) & *Gary Sundin* (SCDNR MRRI)

**11:50-12:30am | Project Webmap Demo and Stakeholders
Feedback with *Caitlyn Bierce* (College of Charleston)**

12:30-2:00pm | Lunch